



Original Research Article

Urine exosome biomarkers of obesity after Lekhana Basti treatment – Report of a pilot study

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A B S T R A C T

Background: Obesity is a rising risk factor for various diseases including cardiovascular diseases and Cancer. The limitations of targeted obesity-treatment approaches employed in the clinic presently underscore the importance of developing integrative management strategies for identification of specific biomarkers of obesity.

Objectives: Given the specificity of exosome/extracellular vesicle (EV) biomarkers, we aimed here to identify the EV biomarkers of Ayurveda treatment – *Lekhana Basti* – for Obesity.

Methodology: A total of eighteen 24-h urine samples from 6 participants with BMI > 30 kg/m² were used in this study, collected over 3 time-points during the *Lekhana basti* (medicated enema for obesity) treatment. Urine EV were isolated using Polyethylene Glycol (PEG). The proteins were resolved by 1-d gel electrophoresis and identified using liquid chromatography coupled with tandem mass spectrometry (LC-MS/MS) and quantified by label-free methods. Significant Protein-Protein Interactions, KEGG pathway analysis and enrichment, functional gene ontology (GO) annotation were identified and shortlisted in comparison to Obesity reference genes from DisGeNET.

Results: With UniProt as a reference subsequent to LC-MS/MS-identification, a total of 210 exosome proteins were identified. Seventy-three proteins were overexpressed in pathway enrichment analysis. Further, GO functional annotation identified 15 common proteins involved. Finally, the 8 hub proteins associated with obesity were identified and their differential expression profile compared between three different time-points during *Lekhana Basti* treatment. Six protein markers overexpressed during obesity were downregulated post *Lekhana Basti* treatment, while 2 markers increased in abundance post-treatment.

Conclusion: To our knowledge, this is the first study to isolate and identify urine EV protein abundance profiles from obese female participants of India. The study results indicate significant changes in the differential expression profile of 8 hub proteins involved in obesity, after *Lekhana Basti* treatment. The biomarker signature of the pilot study indicates the role of Ayurveda treatment and the possible pathways involved in the treatment of Obesity. Further, this study underlines the specificity of urine exosomes/EV as diagnostic markers as well as the potential of Ayurveda treatment in effective management of obesity.

1. Introduction

The World Health Organisation (WHO) defines adults with Body Mass Index (BMI) greater than or equal to 30 kg/m² as obese. The global prevalence of obesity shows a threefold increase from 1975 covering approximately 39% of the global adult population [1]. As per current reports, the prevalence of obesity will increase to 9.5% in men and 13.9% in women by 2040 [1]. Women are much more prone to obesity compared to men [2]. National Family Health Survey (NFHS)-3 of India reports approximately 13% of the women as obese between age 15–59 years [3]. The exponential increase in the risk of diabetes [4], hypertension [5], cardiovascular diseases [6], and cancer [7] makes obesity one of the huge disease burdens.

In the clinic, several options/routes are taken for obesity treatment.

Focus on diet and lifestyle modifications are becoming cornerstones in obesity management [8]. Common treatment strategies include low-calorie diets, intermittent fasting, low-fat diet, calorie restrictions, and some changes in diet style and choices [9]. Some diet and exercise therapies used are of low-, moderate-, intermittent-diet patterns, along with various physician-defined levels of physical activities for a specific duration [10]. Apart from diet modifications, recently, numerous drug therapies of gastric and pancreatic lipase inhibitors are used extensively [11]. Medicinal supplementations including a variety of weight-loss drugs, surgeries, exercise therapies have also been adapted [12]. However, to avoid detrimental as well as undetermined side-effects from these drugs, the regulatory authorities have imposed strict guidelines on its intake [13]. Many drugs which were in the market were also halted after experiencing adverse long-term reactions of treatment [14].

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Dramatically increased interest in surgeries for weight reduction is a recent development in a completely new direction, despite the risks associated with surgical procedures [15]. In addition, development in non-surgical endoscopic weight reduction therapies and evidence-based therapies have more recently gained focus. However, despite the diverse treatment options including diet and lifestyle modifications, surgeries, and drug therapies, the risk and unavailability of a standard obesity management approach has resulted in a huge, yet unmet treatment necessity. Thus, methodologies to understand this commonly seen lifestyle condition and to determine focussed treatment strategies are progressing in a slow but steadfast manner [16,17].

A common diagnostic measure for obesity is evaluating the Body Mass Index of the individual (BMI). This is a surrogate marker for obesity, as it measures only body fatness [21]. BMI values along with waist circumference values have become the standard obesity assessments in clinical practice [22]. However, several studies have shown that in some populations, waist circumference shows only a relative risk of obesity due to ethnicity-specific confounders [9]. Hence, the cut-off ranges of waist measurements in obesity have been stratified for various ethnic groups [23] but still, an effective marker is yet to be identified.

Among various biomarker studies on obesity, exosomes are known as biomarkers of health and disease with high specificity in their identity. Exosome/EV specificity as a biomarker source is well documented with studies focusing on exosome biomarkers of obesity [24–27]. However, most biomarker studies are pre-clinical and controlled, only limited number of proteomic marker studies [28,29] have been conducted in a prospective clinical study to understand the mechanism of biomarker variation during the course of treatment.

In Ayurveda, for the treatment of obesity and dyslipidaemia, a specific type of medicated enema known as *lekhana basti* (lekhana = to scrape) is advised and practised. This type of medicated enema has been practiced since ages, with promising outcomes on obesity symptoms like reduction in weight, BMI and hyperlipidaemia and also a significant change in pro-inflammatory cytokines, immunoglobulins, and functional T cells along and regulation of immunological pathways [18]. Some pre-clinical studies report the effects of basti on hyperlipidemia and its potential activity [19,20]. One of the principles is that it acts on the central nervous system and activates various regulatory pathways via intestinal *trans*-membrane signalling. Despite these developments however, a clear understanding of the mechanism of basti action is still lacking, with only a few research studies systematically addressing this question. An exploration would give researchers a window to recognize and implement lekhana basti as one of the conventional and safe solutions for obesity management, yet highly specific.

Our study specifically focussed on identifying treatment outcomes using medicated enema (Lekhana Basti), an established Ayurveda regimen, in treating obesity. We examined 24-h urine samples collected (i) before, (ii) immediately after treatment and (iii) post-resting period after treatment – for their exosome proteome-level similarities and differences. In doing so, we tested the hypothesis that urine exosome protein content highly specifically represents the respective physiologies producing the urine sample under investigation.

2. Materials and methods

2.1. Chemicals and software tools

Chemicals: Phosphate buffer saline pH 7.4 (HiMedia, Cat no: TL1099), ECP buffer, Bovine Serum Albumin (BSA) (HiMedia, Cat no: MB083), Bradford's reagent (HiMedia, Cat no: ML106), Sodium Dodecyl Sulfate (Merck), Tris base (HiMedia, Cat no: MB029), Acrylamide (GeNei, Cat no: ER13), Bis-acrylamide (GeNei, Cat no: ER14), Ammonium persulphate (GeNei, Cat no: ER11B), TEMED, 2× Laemmli buffer (HiMedia, Cat no: ML021), acetic acid (Loba, Cat no: L0019502500), Comassie Brilliant Blue R-250 (HiMedia, MB153), Acetonitrile,

ammonium bicarbonate, Trypsin, water (HPLC grade, Himedia, Cat no: AS028, GRM1021, RM618, AS077), pre-stained molecular marker (HiMedia, Cat no: MBT092), formic acid, beta-mercaptoethanol (Sigma Aldrich, Cat no: M6250), methanol (Sigma Aldrich, Cat no: 179,337), pH 7, pH 4 and pH 10 calibrators (Eutech, Cat no: FCP62/4.0, 7.0, 10.0) were used.

Consumables and Equipment: Microcentrifuge tubes (Tarsons, Cat no: T500010), pipette tips (Tarsons, Cat no: T521020 & T521010), 15 ml and 50 ml centrifuge tubes (Tarsons, Cat no: T546021 & T546041), pH meter (Hanna Instruments, Cat no: HI98107), SDS plates and comb (BioRad, Cat no: 1,653,311, 1,653,308, 1,653,359), Gel run system (Mini-PROTEAN Tetra Cell, Biorad, Cat no: 1658004EDU), rocking shaker (Rocky max, Tarsons, Cat no: T4080), heating block (Riviera, Cat no: 50,031,001), Power Pac (Bio-Rad PowerPac 300 electrophoresis power supply, Cat no: 68,192–1), vortex shaker (Spinix, Tarsons, Cat no: T3020), mini centrifuge (Spinwin, Tarsons, Cat no: T1010) and Refrigerated high-speed centrifuge (Etek, Cat no: RC4100F) were used for isolation procedures.

Software and database tools: For the data analysis, MSConvert GUI platform, proteogen software tool, Mascot server, The KEGG, DAVID, and Panther pathway analysis tools, STRING protein interaction network tool, DisGeNet database, GO functional annotation database, Cytoscape and Tableau were used.

2.2. Sample collection and processing

A total of eighteen 24-h urine samples (Six each from S1, S2 and S3 timepoints as described in the Clinical Study Details in [Supplementary File 3](#)) from six female participants were randomly selected from the clinical study conducted at IAIM Healthcare, Bangalore on “Effect of Ayurveda based enema on intestinal permeability and endotoxin assay in obesity study” (unpublished work) for analysis (refer to [supplementary file 2](#) for details on the clinical study). The samples were centrifuged and stored at –80 °C until use. The Institutional Ethics committee approval was obtained for this study (TDU/IEC/AL/2017–18/09), registered under the Clinical Trials Registry of India bearing the number CTRI/2018/04/013,354. All the steps followed, from sample collection, processing, EV isolation, proteome data procurement and mining with further downstream analyses are summarized in Figure 1.

2.3. Exosome / EV isolation and LC-MS/MS Quantification

The EV were isolated using standard Polyethylene Glycol-based exosome isolation protocols [30,31]. Bradford's method of protein estimation was performed. Using 12.5% acrylamide gel, SDS-PAGE and the trypsinization was as per standard methods [32]. LC-MS/MS label-free protein quantification [33,34] was done using Thermo Orbitrap nano LC-MS/MS unit [35,36] at 10 ppm peptide mass tolerance, 0.6Da fragment mass tolerance, a maximum of 2 missed cleavages, and a False Discovery Rate (FDR) of <1% with significance threshold $p < 0.05$. The reference sequence was obtained from UniProt (UP000005640).

2.4. LC-MS/MS Data Analysis

The raw files were converted into readable/open file formats (such as. mzML format) using ProteoWizard MSConvert GUI software interface [37]. All the data files were analyzed for protein identification using the Proteogenis QI software, from Waters, MA [38].

During the alignment and quality check, 2 files that did not match the minimum alignment were rejected. A total of 12 files, from 4 subjects were qualified for further analysis. To remove irregularities, the m/z reads only between 10 and 75 min of the run were selectively filtered. As the samples are time-point samples from the same subjects, the input samples were defined into three experimental groups as described earlier. Using the MASCOT server, proteins were identified against reference UniProt Homo sapiens ID (UP000005640). To determine the

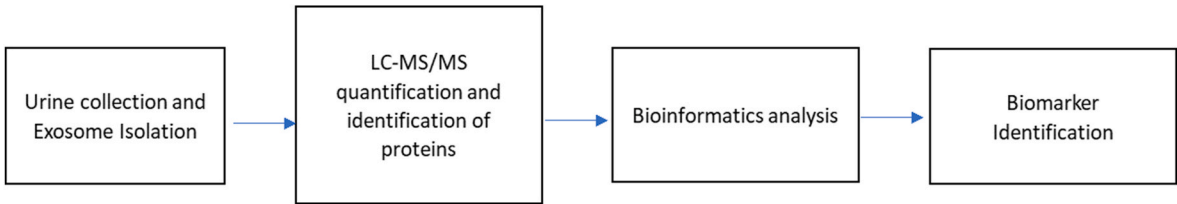


Fig. 1. Flow chart steps followed in this Study.

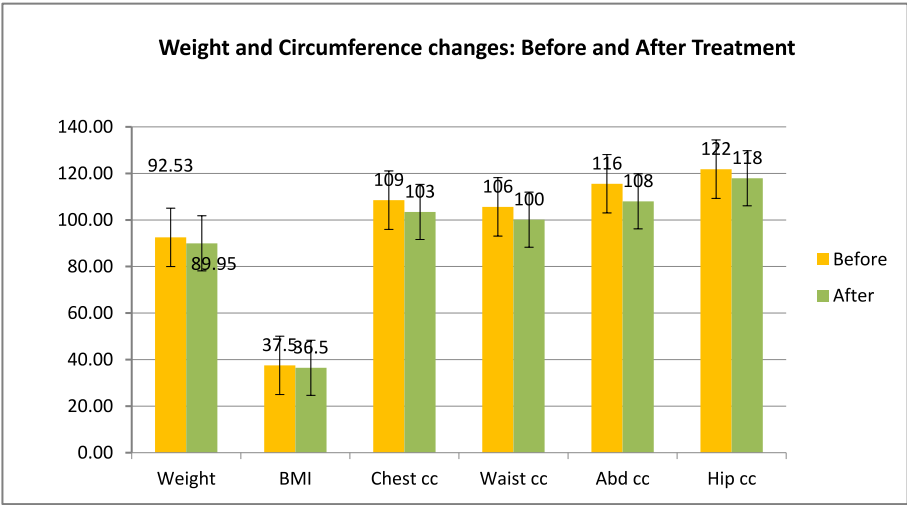


Fig. 2. Anthropometric measurements (Table 1,Supplementary File 2).
cc = circumference change.

correlations between groups, a repeated-measures ANOVA was performed to identify the normalized abundances of the peptides across the time points [39]. (See Supplementary File-1 for full results).

2.5. PPI network

A protein-protein interaction network was built using the online tool STRING, to determine the key protein interactions between the identified proteins. The confidence score for the interactions was set to >0.7 to filter only proteins with strong interactions.

2.6. Pathway analyses and functional annotation

KEGG pathway enrichment analysis was performed using the STRING tool at False Discovery Rate (FDR) < 0.05. Further, the PANTHER tool was used to determine statistical overrepresentation patterns at FDR<0.05. Gene functional annotation analysis was performed using the GO database to identify the key genes involved in the biological processes, molecular functions, and cellular components. The common genes from the PPI network, KEGG pathway analysis and GO analysis were filtered to obtain hub proteins [40,41].

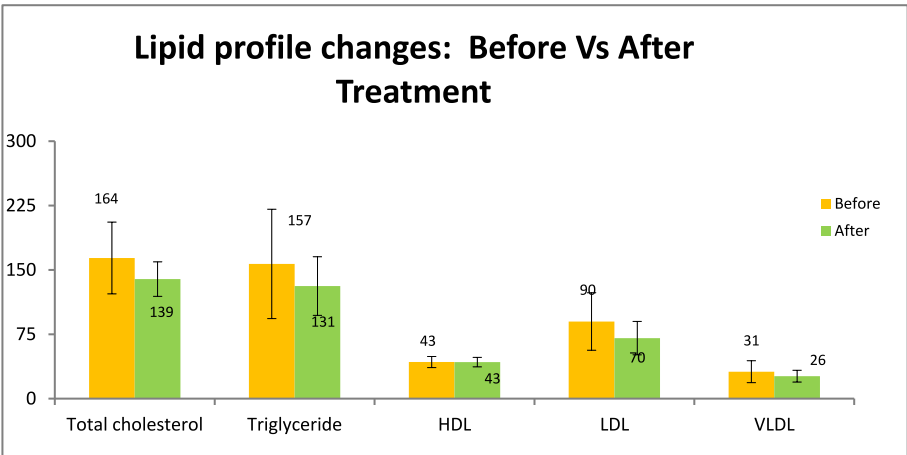
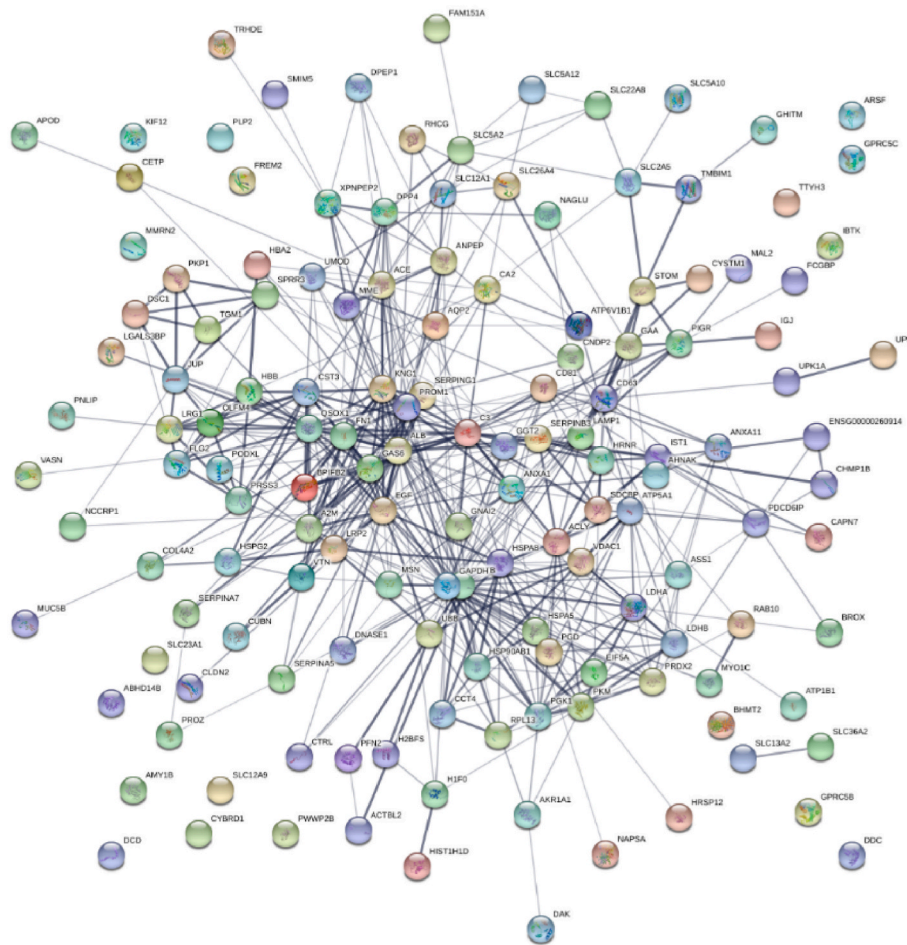


Fig. 3. Lipid profile changes (Table 2, Supplementary File 2).



2.7. Disease-gene relationship

3. Results and discussion

Obesity is becoming a rising concern in the recent past, especially in developing countries [42,43]. The increased energy intake and reduced energy breakdown cause the release of endotoxins, while a considerable increase in intestinal permeability results in low-grade chronic inflammation [44,45]. Integrative management strategies have been implemented to provide effective management lately [46]. Further, the present anthropometric measures have limited scope as markers of obesity at the prognostic level [47–49]. Identification of different

Some biomarker discovery research includes studies on early miRNA markers that have shown a potential for characterizing and modifying obesity and diet patterns [50–52]. Inflammatory markers such as C Reactive Protein, Interleukin-6, Tumor Necrosis Factor Alpha, oxidative stress markers such as lipid oxidation marker F2 isoprostanes, have also been studied as potential markers [53]. Studies also report on gut microbial markers such as *Helicobacter pylori* overgrowth correlating with gastrointestinal hormones to induce low energy breakdown and high energy intakes in obesity [54,55]. WBC, platelet counts, and nutritional markers have also been identified [56]. Metabolomic studies on triglycerides, LDL, HDL, cholesterol levels from plasma, and its functional pathways were shown to have significant associations with obesity and metabolic syndrome [57,58]. Likewise, exosome's specificity as biomarkers of health and disease is well documented, with some studies also focusing on exosome biomarkers of obesity [24]. Exosome/EV component of almost any kind of biofluid is a practical feasibility. Therefore, it has potential utility for cell-specific and focussed prognosis of the targeted cells.

Adipose tissue EV revealed miRNA markers of metabolic diseases such as cardiovascular illnesses, atherosclerosis, diabetes, etc [25]. In an animal model study on plasma EV, an increase in miRNAs (miR-122, miR-192, miR-27a-3p, and miR27b-3p) was seen in lean mice which were transfected with EV from obese mice resulting in glucose intolerance and insulin resistance in the lean mouse, thus showing the action of EV biomarker in metabolic dysregulation [26]. Similarly, a miRNA knockout of adipose tissue EV had around 422 of 653 modifications

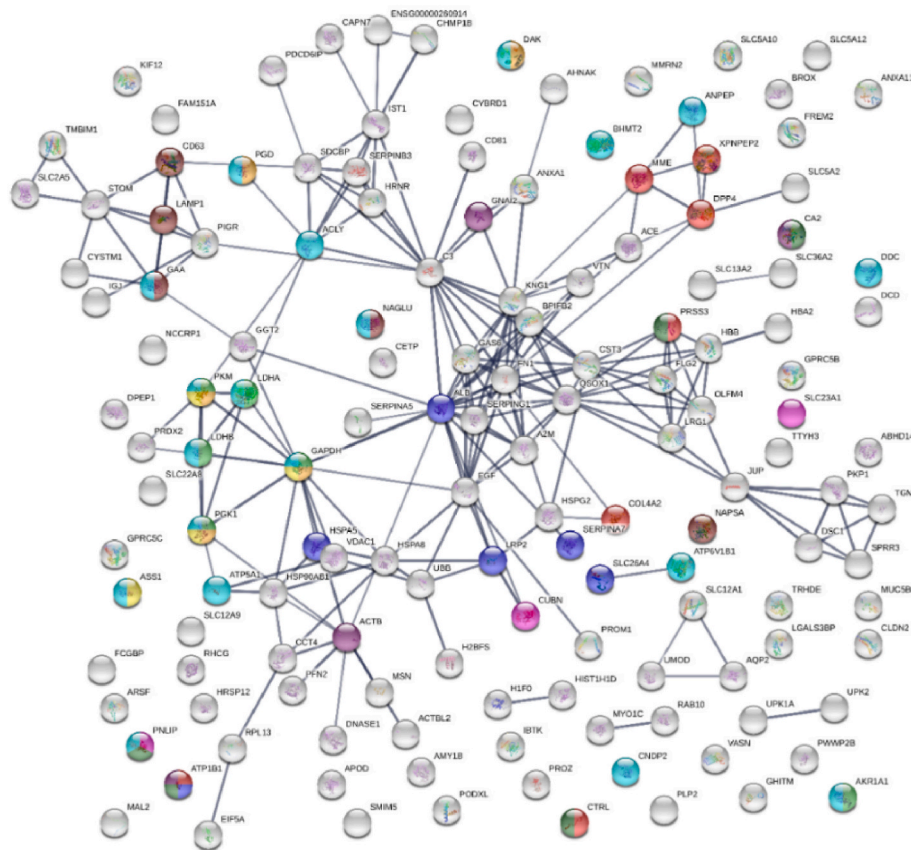


Fig. 5. Protein from enriched pathways in STRING Network.

[27]. However, all the above-said EV biomarker studies are pre-clinical and controlled, only a limited number of proteomic marker studies [28, 29] have been conducted in a prospective clinical study to understand the biomarker during the course of treatment.

Multiple groups have reported adipose-tissue derived EV markers in obesity [26,59,60], in this study, for the first time, we attempted to identify biomarkers of obesity from participants who underwent Ayurveda treatment of using a medicated enema (Lekhana Basti) for restoring gut integrity and reduce obesity [19,20]. We used urine EV as the source of biomarkers to determine the impact of treatment as tracked at the protein expression (biomarker) level.

From LC-MS/MS analysis, a final list of 210 proteins was seen in all the three sample groups, and its respective normalized abundance scores were obtained. From the LC-MS/MS analysis, we identified EV-specific proteins (CD3, CD81, CD63), which validates the isolation and purification method used in the study [61]. Further, the STRING protein-protein interaction network provided a strong interaction network at the confidence of >0.7 (See Fig. 4).

Obesity is seen with dysregulation of glucose, protein, vitamin, and immune pathways such as upregulation of cell adhesion, transmembrane migration activities, and increased amino acid, protein and glucose regulation pathways, upregulation in circulatory immune and inflammatory process, enhanced matrix part, protein, peptidase, extracellular matrix bindings [62–64]. In our study, KEGG pathway enrichment analysis showed 12 enriched pathways (73 proteins) related to metabolic dysregulations where 19 proteins were mapped to the dysregulated metabolisms, 7 proteins were responsible for protein digestion and regulation, followed by upregulation of proteins associated with glycolysis and gluconeogenesis, pancreatic secretion, and gastric acid secretion. Proteins from the KEGG analysis are highlighted in Fig. 5 of the string interaction network.

Further, from Panther over-representation analysis, 31 molecular

pathways were overexpressed at >11 times fold enrichment. The top enrichment was observed in metalloproteinase (MMP) activity at >100 fold change. MMPs are proteinases associated with functioning in matrix degradation, adipogenesis, and differentiation in obesity and cardiovascular conditions [65,66]. Our results strongly suggest that the significant rise in MMP activities directly correlates with obesity. Most of the overrepresented pathways from our results were associated with upregulation in intestinal transmembrane activity including dipeptidyl-peptidase activity, sugar transmembrane transporter, hexose, glucose, carbohydrate transmembrane activities as seen in obesity [67–70].

From the GO functional annotation, a total of 25 proteins were found common among Biological, Molecular, and Cellular functions shown in panels a, b, c respectively of Fig. 6. The proteins involved in the peptidase regular and inhibitor activity were over-expressed in biological processes, ficolin-1-granule was high in cellular functions, and negative regulation of peptidase activity, peptidase inhibitor activity, and endopeptidase activities were the major molecular functions significantly affected, as indicated in the literature. Interestingly, many of the common proteins associated with obesity were mapped to cancer and metabolic diseases as reported in many studies [70–75].

Utilizing the intersection of PPI, KEGG, and GO, we obtained an intersection list of protein that were mapped to reference genes from DisGeNET. This was used to further shortlist the hub proteins involved in obesity. We found 8 hub proteins related to obesity (HSPA5, C3, GAPDH, ACTB, ALB, DPP4, ANPEP, and CTRL) represented in Fig. 7. and discussed in the following section.

Thus 8 biomarkers showed a differential expression pattern that is similar to signatures of obesity. Studies from other groups show that an increased interaction of Reticulon-3 with Heat-Shock Protein 5 (HSPA5) activates Sterol regulatory element-binding protein 1 (SREBP-1) which induces obesity and accumulation of triglycerides [76]. Increased Serum

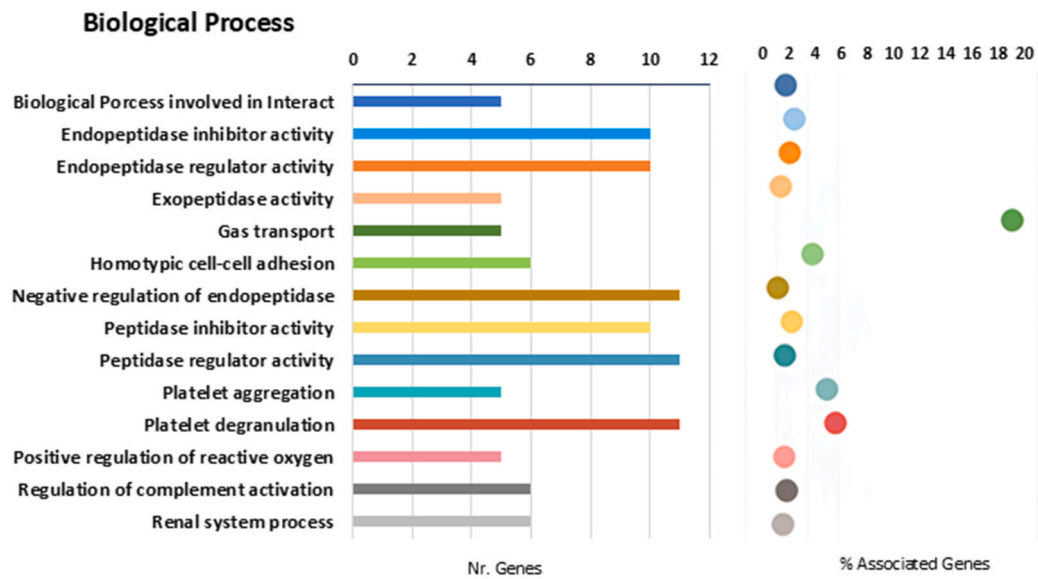


Figure a

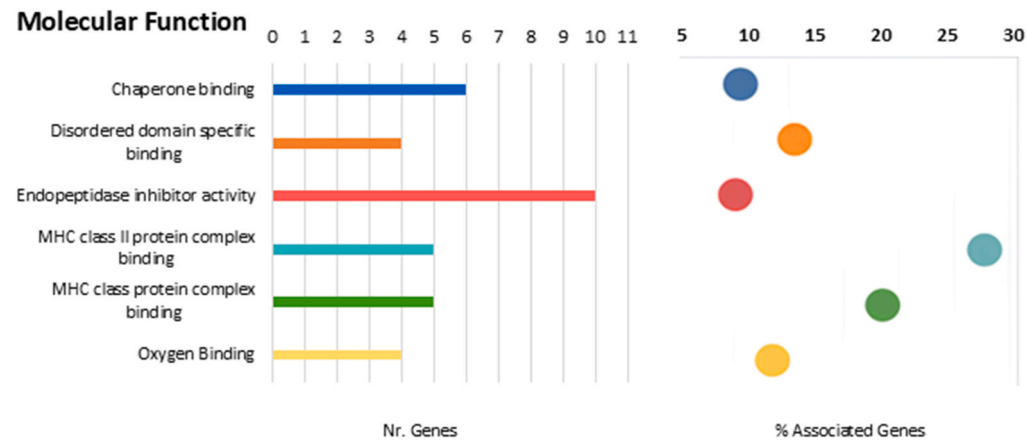


Figure b

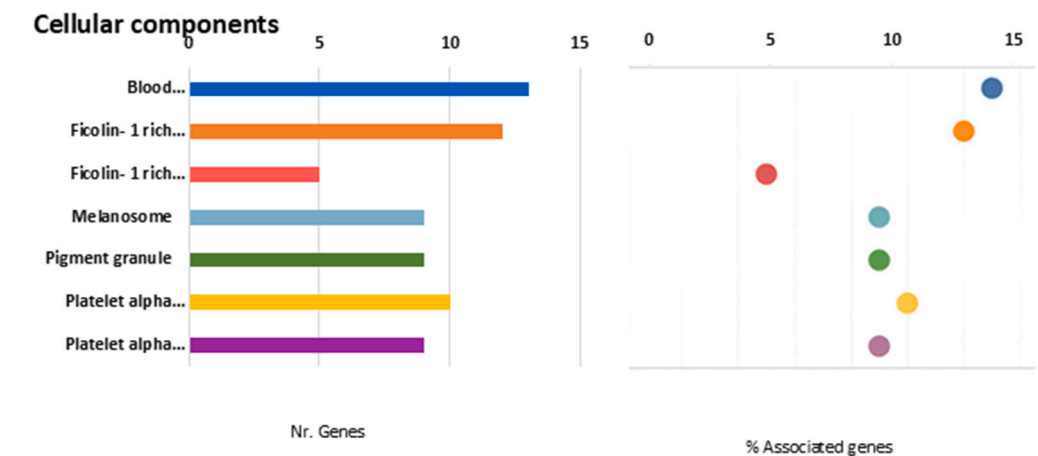


Figure c

Fig. 6. a: GO annotation of proteins involved in Biological Process
b: GO annotation of proteins involved in Molecular Functions
c: GO annotation of proteins involved in Cellular Components.

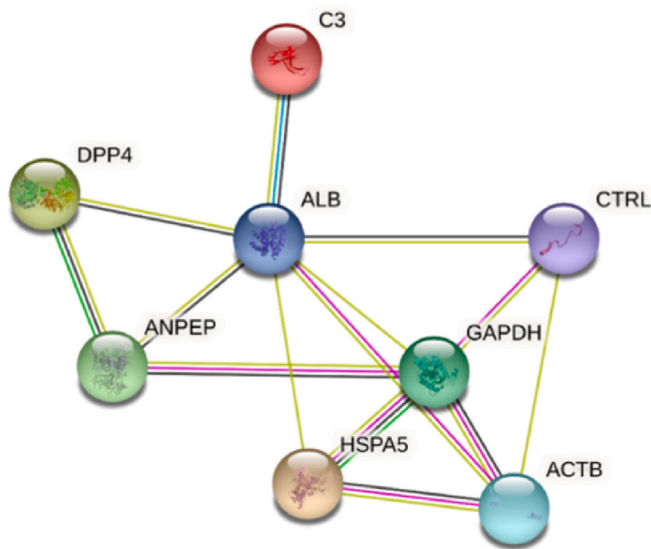


Fig. 7. Hub protein interaction of the exosome biomarkers.

Complement (C3) shows significant relation to adipose tissue variations in obese conditions [77,78]. Glyceraldehyde 3-phosphate dehydrogenase (GAPDH), one of the housekeeping genes which plays a major role in glucose and nuclear functions, is overexpressed under obese conditions, affecting major metabolic pathways [79–82]. Similarly increased expression of Serum Albumin (ALB), an indicator leading to morbid obesity [83], and under-expression of Actin (ACTB), is seen in obese conditions [81]. Dipeptidyl peptidase-4 (DPP4), an adipokine made in the adipose tissues, is positively related to obesity [68]. Similarly, Alanine aminopeptidase (ANPEP), encodes metalloprotease, a highly glycosylated protein that increases with increased adipogenesis in obesity [84]. Finally, Chymotrypsin (CTRL), is a protease inhibitor required for insulin metabolism is downregulated in obesity [85]. In this study, the protein expression data shows a ‘correction pattern’ of certain biomarkers from the pre-post treatment analysis. HSPA5, C3, GAPDH, ALB, DPP4, and ANPEP, which are overexpressed in obesity, were reduced post-treatment. Conversely, CTRL and ACTB which are under-expressed under obese conditions had increased expression levels post-treatment, indicating a general “dysregulation-reversal” – irrespective of the directionality of dysregulation.

Some of the limitations of this study include the following: Firstly, it is based on a small size ($n = 6$); when studied in a larger study sample size context, some of these specific urine exosome biomarkers of obesity-treatment may get re-validated. Next, an investigation of holistic treatment methods into underlying mechanisms still needs to be addressed on a bigger scale with a diverse (with respect to gender, ethnicity background, age and additional co-factors that may be confounding in small size contexts) and robust study design. Further, future studies can help identify clinical diagnostic exosome markers and mechanisms of action of holistic treatments such as Ayurveda. To address the limitation of our study, a future investigation involving non-proteomic validation methods such as ELISA/Western blot would be a fundamental requirement - for the proteins that floated to the top of the biomarker candidate list, namely HSPA5, C3, GAPDH, ALB, DPP4, and ANPEP. This would potentially validate the urine exosome biomarkers of obesity/treatment borne out in this study.

4. Conclusion

The complex pathophysiology of obesity has rendered both its management and treatment, a formidable challenge. Given our experience with the high specificity of urine exosome/EV in markers of different disease phenotypes, we explored exosome/EV markers of

obesity treatment in this study. The study outcomes, pathway enrichment, and functional annotation indicated dysregulated metabolic pathways and functions closely associated with obesity. The abundance of eight dysregulated hub genes representing urine EV biomarkers related to obesity was reduced in the post-treatment condition. This is a reliable indicator of the potential of Ayurveda based medicated enema (Lekhana Basti) in correcting the metabolic dysfunctions, altering the intestinal permeability and restoring gut dysbiosis. This is the first study to have identified urine exosome biomarkers of obesity and its differential expression changes after Ayurvedic treatment in Indian female subjects. In summary, this study supports the need and acceptance of integrative health approaches to effectively address obesity-questions in the clinic.

Data availability

The dataset supporting the findings of this paper belongs to the TDU, Bangalore, India. Currently, the data are not publicly available but can be obtained upon reasonable request and with permission from the corresponding author, Satish P Ramachandra Rao.

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Declaration of generative AI in scientific writing

No generative AI tools were used in the preparation of this manuscript. All content was created and written by the authors.

Author contributions

PD conceptualised and conducted clinical study, SPR conceptualised proteomics study and BV conducted proteomics study under the guidance of SPR, BV and SPR did the proteomic data analysis BV, SPR and PD drafted, reviewed and edited the manuscript, GS reviewed and edited the manuscript for final submission.

Conflict of interest

The authors declare that there are no conflicts of interest regarding this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jaim.2024.101043>.

Supplementary Materials

Supplementary File 1 Summarizes Details of the expression profile of urine EV proteins.

Supplementary File 2 Summarizes Anthropometric measurements and lipid profile.

Supplementary File3 Summarizes clinical details.

Supplementary File 4 AResponse to reviewers comments (Reveiw #6 comment no. 6).

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